

CFB BORDEN, ON, Canada

WMO: 715340

Lat: 44.270N

Lon: 79.910W

Elev: 730

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-4.8	-19.7	1.9	-10.5	-11.8	3.0	-3.4	24.9	25.2	22.5	27.2	3.0	340	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	89.1	71.9	85.6	70.1	82.8	68.5	74.1	84.7	72.4	82.3	70.8	79.5	12.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.9	117.1	78.8	69.3	110.5	77.4	67.6	104.0	75.8	38.1	84.6	36.6	82.3	35.1	79.8	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	18.8	16.7	-25.8	93.7	5.6	2.8	-29.8	95.7	-33.0	97.3	-36.1	98.9	-40.2	100.9
			-25.6	77.1	6.1	1.7	-29.9	78.4	-33.5	79.3	-36.9	80.3	-41.3	81.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	18.9	19.5	28.8	40.9	54.0	63.1	68.0	66.5	59.7	47.5	36.5	25.7		
	DBStd	19.80	12.91	11.46	11.36	9.35	9.11	7.33	6.25	5.91	8.04	8.53	9.37	10.26		
	HDD50	4166	963	853	661	303	58	1	0	0	10	152	412	753		
	HDD65	7939	1428	1273	1121	723	364	122	39	52	200	545	854	1218		
	CDD50	2065	0	0	4	30	182	395	559	512	300	76	7	0		
	CDD65	365	0	0	0	1	22	65	134	99	41	3	0	0		
	CDH74	4271	0	0	3	34	323	828	1492	1064	484	43	0	0		
	CDH80	1314	0	0	0	6	96	277	483	305	142	5	0	0		
	WSAvg	7.2	8.1	8.0	7.8	8.2	7.2	6.5	6.0	5.7	6.0	7.1	7.5	7.8		
	PrecAvg	32.8	2.4	2.0	2.2	2.6	2.9	3.1	3.2	3.0	3.0	2.8	2.9	2.7		
(15) Precipitation	PrecMax	41.8	4.1	4.2	5.7	5.2	5.2	8.1	4.7	6.6	5.3	5.3	5.4	4.3		
	PrecMin	24.5	1.1	1.0	0.7	0.9	1.0	0.8	1.3	1.4	1.2	0.9	1.0	1.2		
	PrecStd	3.9	0.8	0.9	1.1	1.1	1.0	1.6	1.1	1.1	1.0	1.1	1.1	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.7	50.8	67.2	79.3	87.3	91.5	93.0	91.9	89.5	80.0	64.4	53.5		
		MCWB	46.8	45.8	58.1	60.9	68.3	72.7	74.7	72.4	72.1	66.0	54.7	51.1		
	2%	DB	43.0	43.7	57.0	70.4	82.6	87.1	88.8	86.7	84.1	73.0	59.0	46.8		
		MCWB	40.1	40.7	49.9	56.9	66.5	70.4	71.8	71.1	69.8	62.8	53.2	44.0		
	5%	DB	39.2	38.6	50.6	64.2	76.9	82.8	85.1	83.2	79.9	66.9	55.2	42.0		
		MCWB	36.8	35.3	44.3	52.7	63.1	68.3	70.0	69.4	67.4	59.5	50.3	39.4		
	10%	DB	35.8	35.2	44.8	58.3	72.2	78.6	81.9	80.0	75.2	62.1	50.7	38.4		
		MCWB	33.9	32.6	39.7	48.7	59.5	66.1	68.2	67.7	65.0	56.3	46.8	36.3		
	0.4%	WB	47.6	46.8	59.5	63.8	71.3	74.9	76.6	76.4	74.5	69.4	57.6	51.4		
		MCDB	48.7	48.9	65.9	74.4	82.4	86.7	88.4	86.1	84.1	76.2	60.8	53.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.0	41.1	50.9	58.8	68.3	72.6	74.2	73.3	72.0	64.1	54.3	44.3		
		MCDB	42.4	44.0	55.3	68.7	79.8	84.6	84.9	83.1	80.6	70.1	58.5	46.0		
	5%	WB	36.8	35.7	45.4	54.2	65.3	70.2	72.3	71.5	69.3	60.7	50.5	39.9		
		MCDB	38.7	37.8	50.1	61.8	74.2	79.8	81.7	80.2	76.6	65.6	54.3	41.8		
	10%	WB	33.9	32.6	40.0	49.6	61.6	67.7	70.5	69.5	66.7	57.2	47.1	36.3		
		MCDB	35.6	35.1	44.1	57.5	69.8	75.9	79.1	76.9	73.2	61.4	50.8	38.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.6	19.0	21.9	25.4	23.9	23.9	23.7	24.2	18.7	16.1	13.5		
		MCDBR	14.9	19.2	26.8	33.1	31.3	29.1	27.8	28.1	29.0	27.3	22.4	15.6		
		MCWBR	13.5	16.8	19.6	20.9	17.1	15.0	13.5	13.9	15.5	17.5	17.4	13.7		
		MCDBR	14.3	17.6	24.9	28.1	27.6	25.1	23.5	22.8	24.0	23.4	20.4	15.4		
	5% WB	MCWBR	13.7	16.2	19.2	19.7	15.9	14.0	12.2	12.4	13.9	15.9	16.8	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.301	0.318	0.357	0.387	0.410	0.404	0.396	0.364	0.345	0.310	0.283		
	taud		2.363	2.292	2.401	2.349	2.317	2.275	2.311	2.319	2.408	2.446	2.490	2.425		
	Ebn at Noon		267	281	292	287	280	273	273	272	272	263	256	256		
	Edh at Noon		24	31	32	37	40	42	40	38	32	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		427	706	1118	1422	1710	1869	1886	1624	1282	751	447	311		
	RadStd		49	87	117	142	171	136	122	85	121	78	63	38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page